

FANUC Robot **series**

R-30iB AND R-30iB PLUS CONTROLLER

Spot Easy Normal

OPERATOR'S MANUAL

Version 8.10 and later

MAROBENOR11121E REV. D

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1

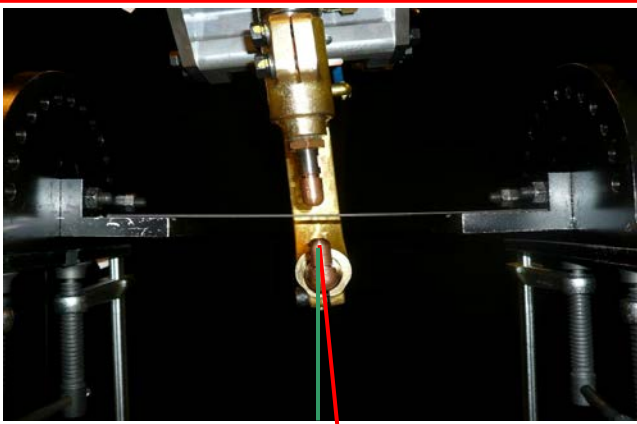
OVERVIEW

1. OVERVIEW

Orientation error (gun not normal to panel) results in uneven force distribution and ultimately uneven current density at spot. Spot Easy Normal (R768), also known as Servo Gun Normalization, is a utility that determines proper orientation of gun to the welding surface. Easy Normal automatically detects panel orientation, and makes gun tip normal to the panel, with no operator intervention.

Result after normalization for a panel with orientation $W=0, P=0$, is shown below.

Before Normalization (error = 5degrees)

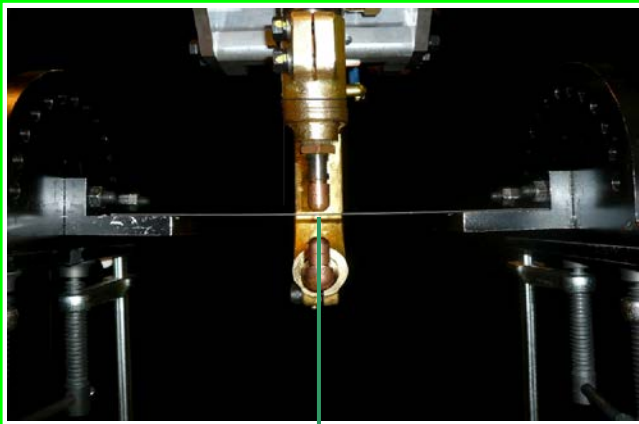


Position screen - Before normalization

Group1:	
POSITION	
World	Tool: 1
Configuration: F U T, 0, 0, 0	
x: 2400.000	y: 190.000 z: 160.000
w: 5.000	p: 5.000 r: 50.000

Group2:	
POSITION	
Joint	Tool: 1
J1: -55.000	

After Normalization (error ~0 degrees)



Position screen - After normalization

Group1:	
POSITION	
World	Tool: 1
Configuration: F U T, 0, 0, 0	
x: 2405.804	y: 190.535 z: 159.932
w: 0.534	p: -0.050 r: 180.000

Group2:	
POSITION	
Joint	Tool: 1
J1: -40.140	

location correction

rotation correction

gun tip correction

The normalization function does not move the fixed tip to the panel and does not touchup the point. Therefore, when normalization function completes, either touchup the point manually or press F4 Auto Touchup to actually record new orientation.

1.1. Limitations

- Dual gun is not supported
- Scissor-type (no fixed tip) gun is not supported
- Pedestal gun not supported
- Normalization is performed relative to gun tip. I.E. is assumed that surface orientation of lower panel is the same as upper panel.

2 SETUP

2. SETUP

The following items must be completed before the Spot Easy normal can be used:

- 1 Basic Servo Gun setup - refer to the *Servo Gun Function Operator's Manual*.
 - master + calibrate Servo Gun axes
 - autotune
 - calibrate pressure
 - setup utool
- 2 Deflection calibration – refer to the *Servo Gun Function Operator's Manual*.
- 3 Torque limit calibration – refer to Section 2.1.
- 4 Tool directions – refer to Section 2.2.

2.1. Torque Limit Calibration

Torque limit calibration must be performed for

1. initial setup
2. Before starting normalization if gun characteristics are known to change significantly with operating conditions (hot / cold gun, etc).

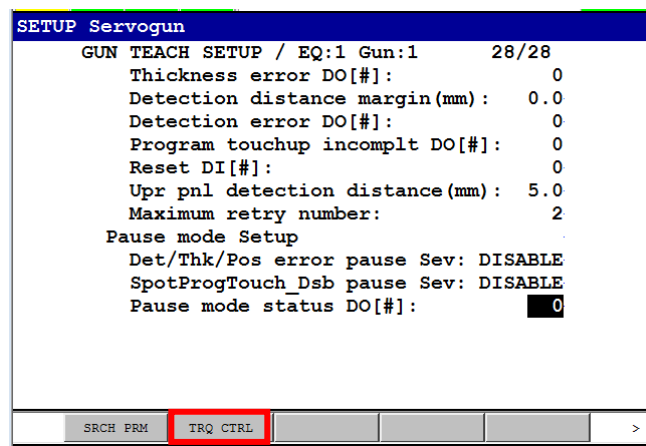
note: torque limit calibration is automatically performed during autotuning.

Normally, it is not necessary to perform procedure 2-1.

Procedure 2-1 Torque Limit Calibration

Step

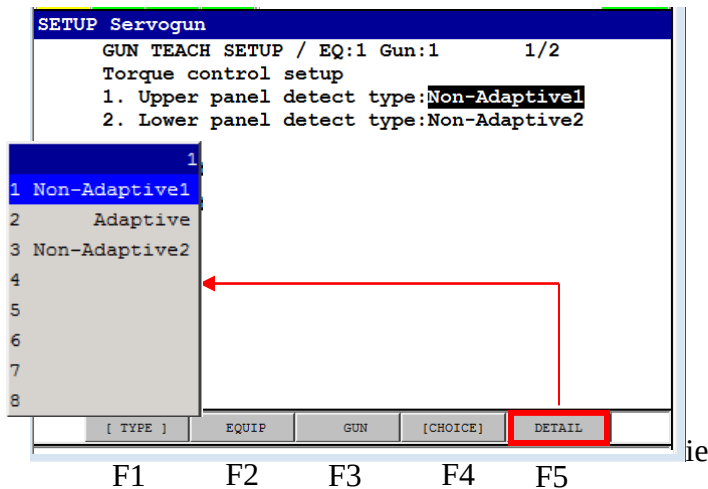
- 1 Go to gun teach setup screen.



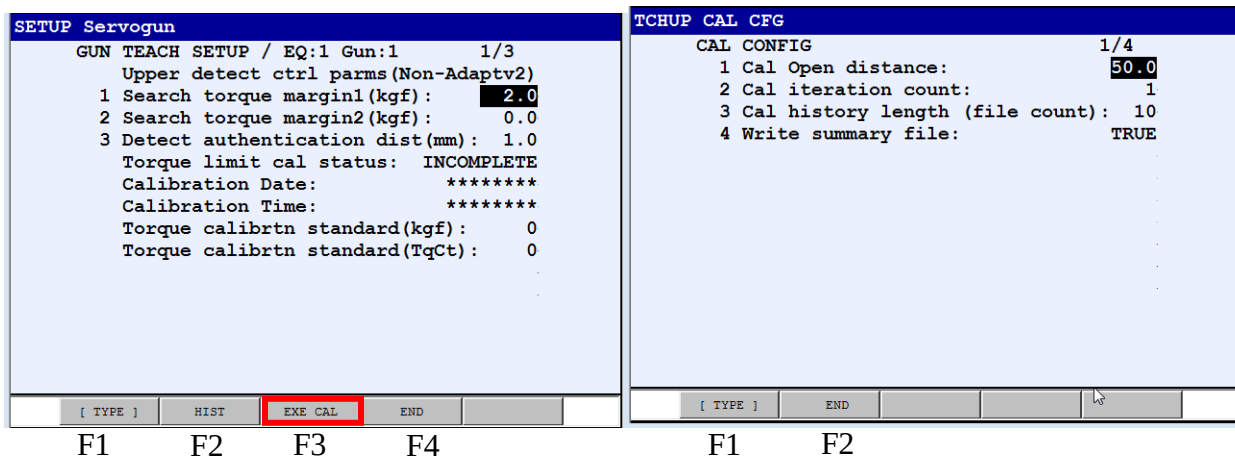
F1

F2

- 2 Press Next, then F2: TRQ CTRL to display torque control setup screen



- 3 Press F4: CHOICE to change upper panel detect type to Non-Adaptive2 (if it is set to Non-Adaptive1 or Adaptive)
- 4 Press F5: DETAIL, to display upper detect ctrl parameters (below)



- 5 Press Shift + F3 to calibrate torque limit. Configuration settings for torque limit calibration can be displayed and/or modified by pressing F3 without shift. This screen is shown above on right.

2.2. Tool direction identification

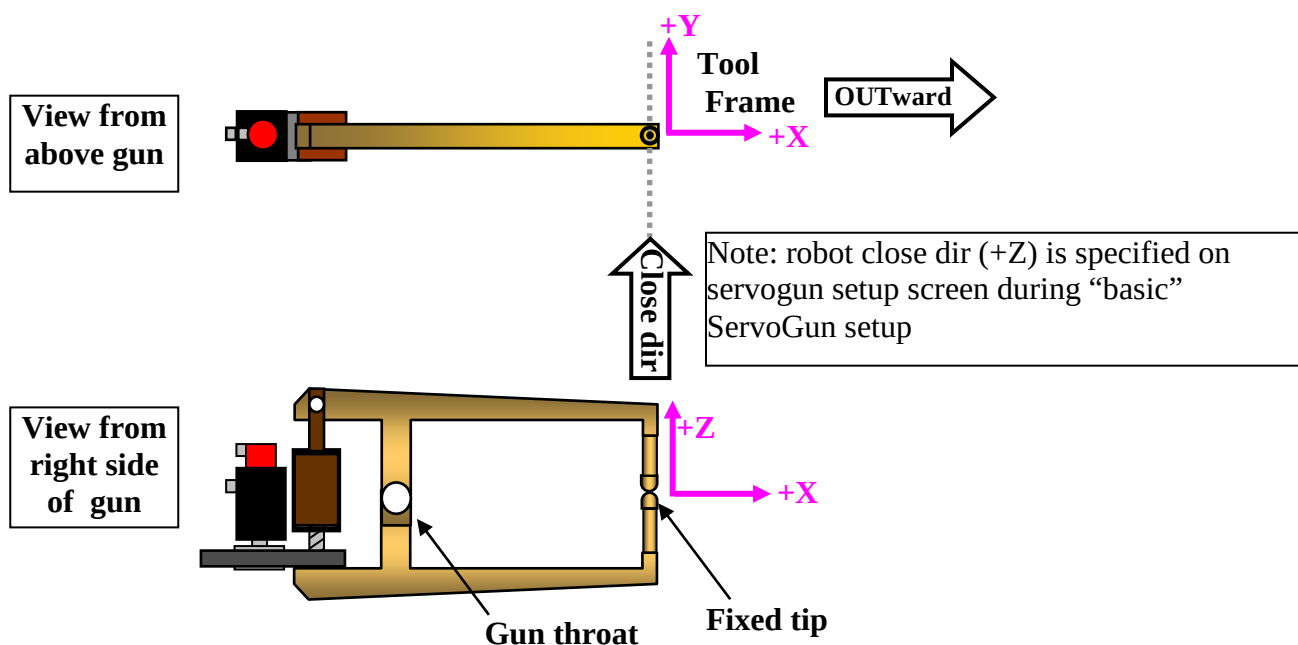
Outward direction is defined as direction from gun throat to weld tip (cap). Specify outward direction in accordance with utool definition. Outward direction is +x, in the diagram below.

Set outward direction via Item 8 on normalize touchup config screen. Refer to next section for more details on normalize config screen.

SETUP Servogun	
NORMALIZE TOUCHUP CFG / EQ:1 Gun:1	
1 Shape type:	Diamond
2 Shape radius(mm):	5
3 Shape speed(mm/sec):	200
4 Search speed(mm/sec):	4
5 Rotation detection tol(deg):	10.0
6 Rotation correction tol(deg):	10.0
7 Normalize guntip distance(mm):	5.0
8 Outward direction:	+X

F4: Choice

- 1 +X
- 2 -X
- 3 +Y
- 4 -Y
- 5 +Z
- 6 -Z



WARNING

If outward direction or robot close direction are set incorrectly, rotation will be incorrect and collision can occur. Be sure that outward direction and robot close direction are correct.



CAUTION

utool error degrades normalization accuracy. Confirm that utool is accurate.

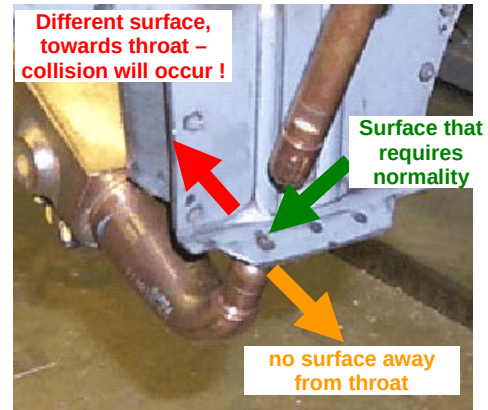
3

CONFIGURATION

3. CONFIGURATION

The robot performs a search to “find” the surface. The searchable envelope is finite and spot location-dependent. Sometimes, the surface around a desired spot location may change or may not exist.

Additionally, the robot arm may collide with another object in cell when moving around the desired spot location to “find” the surface. These issues are illustrated in image on right.



3.1. Easy Normal Configuration

Consider these issues, then use normalize touchup config to optimize pattern and radius for search. Normalization configuration settings (values) are persistent, so values don't have to be re-entered at each spot. However, new values should be entered if values needed for current spot differ from previously entered values.

Procedure 3-1 Configure normalization settings at a spot

Step

- 1 Go to normalize touchup configuration screen. Either:
 - a Go to editor screen and press F2 (without shift) OR
 - b Go to teach setup screen, and press F3 (without shift)
- 2 Modify settings as required for current spot.
- 3 Press PREV, END or EDIT to exit normalize config screen.

SETUP Servogun		
NORMALIZE TOUCHUP CFG / EQ:1 Gun:1		
1	Shape type:	Diamond
2	Shape radius(mm):	5
3	Shape speed(mm/sec):	200
4	Search speed(mm/sec):	4
5	Rotation detection tol(deg):	10.0
6	Rotation correction tol(deg):	10.0
7	Normalize guntip distance(mm):	5.0
8	Outward direction:	+X

F4: Choice

- | |
|----------------------------|
| 1 Diamond |
| 2 Square |
| 3 Triangle pointed outward |
| 4 Triangle pointed inward |
| 5 Line pointed inward |
| 6 Line pointed leftward |

Table 3-1 Normalize config screen parameters

ITEM	DESCRIPTION
Shape type	Shape and pattern for search
Shape radius(mm)	Distance from original point to test point
Shape speed(mm/sec)	Speed between test points (lateral motion over panel)
Search speed(mm/sec)	Speed for panel detection (motion in gun close direction)
Rotation detection tol (degrees)	Worst case error (angle) expected between current orientation and actual orientation. This parameter determines clearance for gun and robot during motion to test points.
Rotation correction tol (degrees)	Maximum rotation correction applied without user consent. If rotation change is less than rotation correction tol (within expected range), rotation correction is applied automatically. If measured rotation error is larger than rotation correction tol, user must respond yes to prompt before rotation correction occurs.
Normalize distance margin(mm)	Clearance for variation and utool error at guntip.
Outward direction	Direction that corresponds to “out” direction. Note that value is ** (unit) by default.

See Section 3.1.1 for more for details on shape parameters.

3.1.1 Shape Type and Shape Radius

Typically, square or diamond shape with a small radius is suitable at most spot points. However, this shape and/or radius may not be suitable at ALL points – no surface may exist at some test points (within shape). Radius and/or shape must be set differently at these spot points.

Consider the following guideline, when selecting shape and/or radius:

Test points MUST provide

- obstruction-free motion
- same orientation on both sides (surfaces) of panel

Test points SHOULD provide

- maximum separation, for best accuracy

Six different shapes are supported. All shapes include a center test point (point where gun tip touches the panel before normalization). The Default shape is diamond. The route direction is clockwise for diamond, square and triangle.

Figure 3-1 Test point patterns for 2 angle correction (corrects 'w' & 'p' angles)

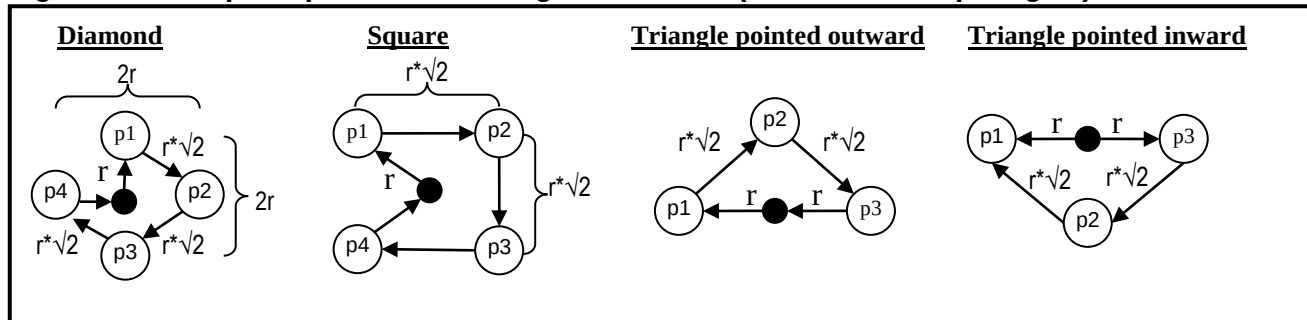
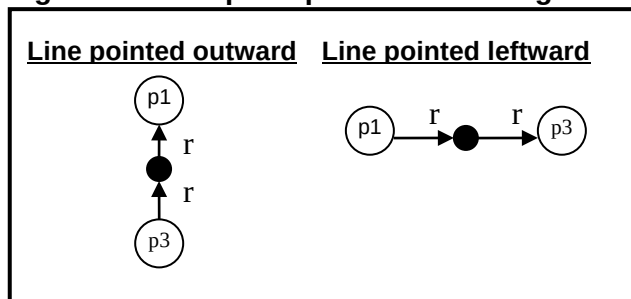


Figure 3-2 Test point patterns for 1 angle correction (correct w OR p, but not both)



Legend

- original point (testpt0)
- p# test point (testpt1-N)
- r search radius

The default radius = 5mm, however the radius can be set as small as 1mm.

However, the difference between test point measurements becomes very small if a radius like 1mm is specified. A radius $\geq 3\text{mm}$ is recommended.

3.1.1.1. Shape speed

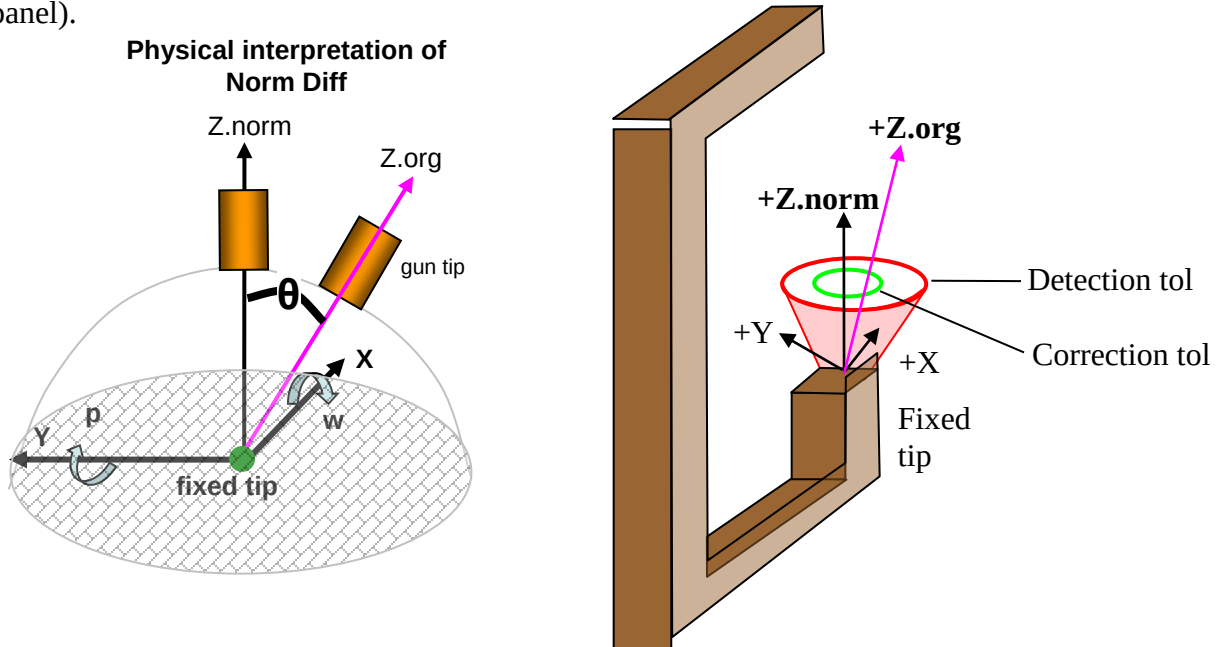
Shape speed is speed of robot between test points (aircut speed). Shape speed is clamped to 250mm/sec and is independent of override.

3.1.1.2. Search speed

Search speed is panel detection speed (gun close speed) at each test point. Search speed is also independent of override. Default speed is 4mm/sec. Slower speed may improve accuracy of result, but also increases execution time for normalization process.

3.1.1.3. Rotation detection tolerance and rotation correction tolerance

Norm Diff(θ) is the angle between Z.org (initial vector out of fixed tip) and Z.norm (vector normal to panel).



4

EXECUTION SEQUENCE

4. EXECUTION SEQUENCE

4.1. Procedure

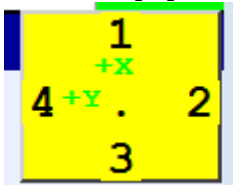
- 1 Move the robot to the spot instruction where normalization is required. The gun tip and robot tip must not be pressing into the panel.
- 2 Go to 2nd fkey page by pressing the “NEXT” hard key, and find “F3 Normaliz”.
- 3 Press “F3 Normaliz” to begin the sequence.

```
SPOTPROG
1:L P[2] 2000mm/sec
  SPOT[SD=1, P=1, S=1, ED=1]
|INST |REL_VIEW| NORMALIZ | | EDCMD |
```

F1 F2 F3 F4 F5

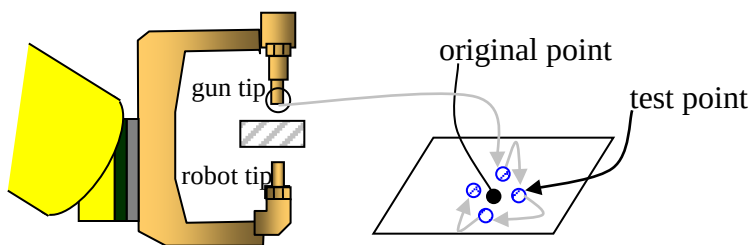
Normalization process starts.

Popup window appears



- 4 Search process executes:
 - a. Gun tip touches panel at the original position
 - b. Robot moves through the predefined test pattern around the original point

Plane of panel is found from z location at multiple points around cap. It is assumed that surface around cap has same orientation as surface under cap.



- 5 Search process completes
 - When the test point search pattern is complete and the computed angle is within the specified tolerance, the robot will move so that it is normal to the part.
 - If the computed angle is not within the specified tolerance, the user will be prompted before moving to the normalized position.

4.2. Orientation compare function

After the normalization function is complete, the user can easily compare the original orientation to the normal orientation by moving the robot between these orientation locations.

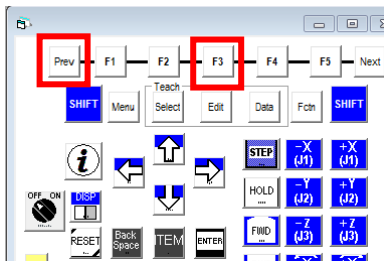
The orientation compare function – pressing PREV and F3 key – changes orientation as follows:

- . if curpos == NORM, then robot & guntip move to org pos
- . if curpos == ORG, then robot & guntip move to NORMALIZE pos

However, if normalization has NOT been performed or if the robot has been jogged away from the original position then an appropriate error message is displayed.

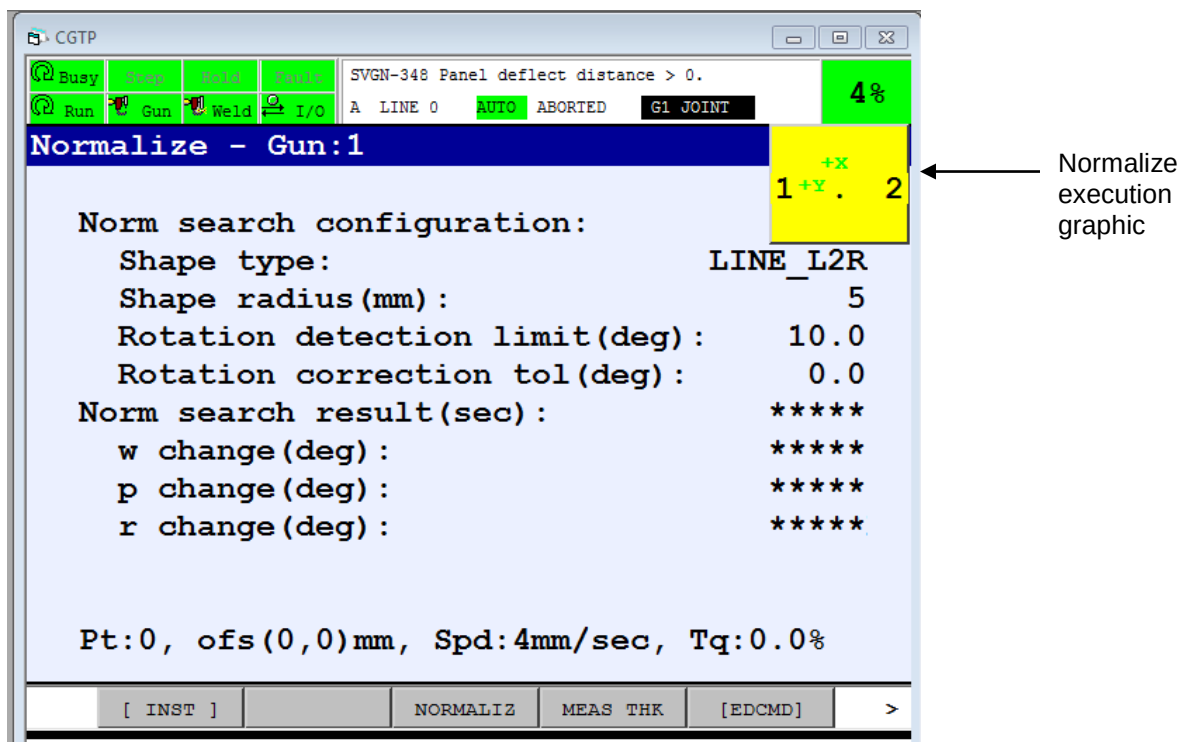
PREV + **F3**

NOTE: The prev. key is used because it is adjacent to shift key. I.E. it is easy / convenient to execute normalization via Shift + F3, then compare positions by pressing Prev + F3.

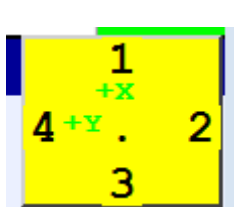


4.3. Normalize execution graphic (popup window)

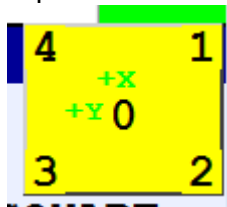
In addition to the summary screen, a yellow popup window is displayed during normalization process. The location of the symbols (number or *), correspond to the test point pattern. The numeric value indicates test point execution sequence(order). I.E, 1=testpt1,2=testpt2, etc. Solid (non-blinking) numbers indicate points where search is incomplete. A blinking number indicates a test point where search is in progress. * indicates a test point where search has been completed.



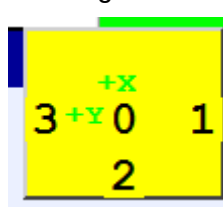
Diamond



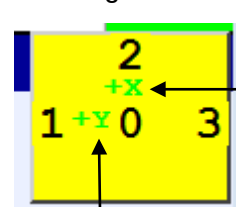
Square



TriangleIn



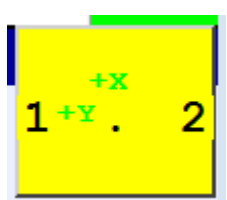
Triangle Out



Line Out



Line Left

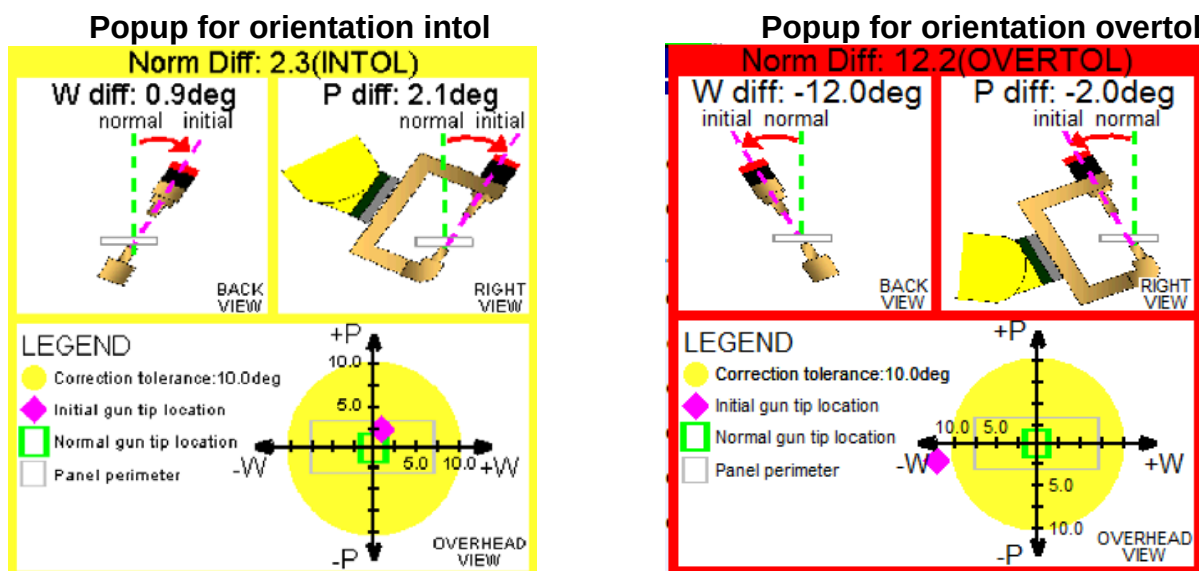


Note:

Outdir and leftdir axis are identified in green text, at test point0. Outdir and leftdir are not displayed on subsequent test points.

4.4. Normalize result graphic (popup window)

At normalize completion, a result graphic displays. This window contains a graphic/visual to clarify the sense/direction of the angle's polarity.



NOTES:

- Norm Diff is difference between Z.org and Z.norm
- If orientation is in tolerance (yellow circle), then background is yellow
- If orientation is NOT in tolerance, then background is red
- Previous normalization result can be displayed by pressing SHIFT + Prev
- Press reset to cancel popup window
- Set \$SGNRMCFG.\$SHO_OVRHDVW = FALSE, to hide overhead view (default=TRUE).
- Popup persistence is specified in table below

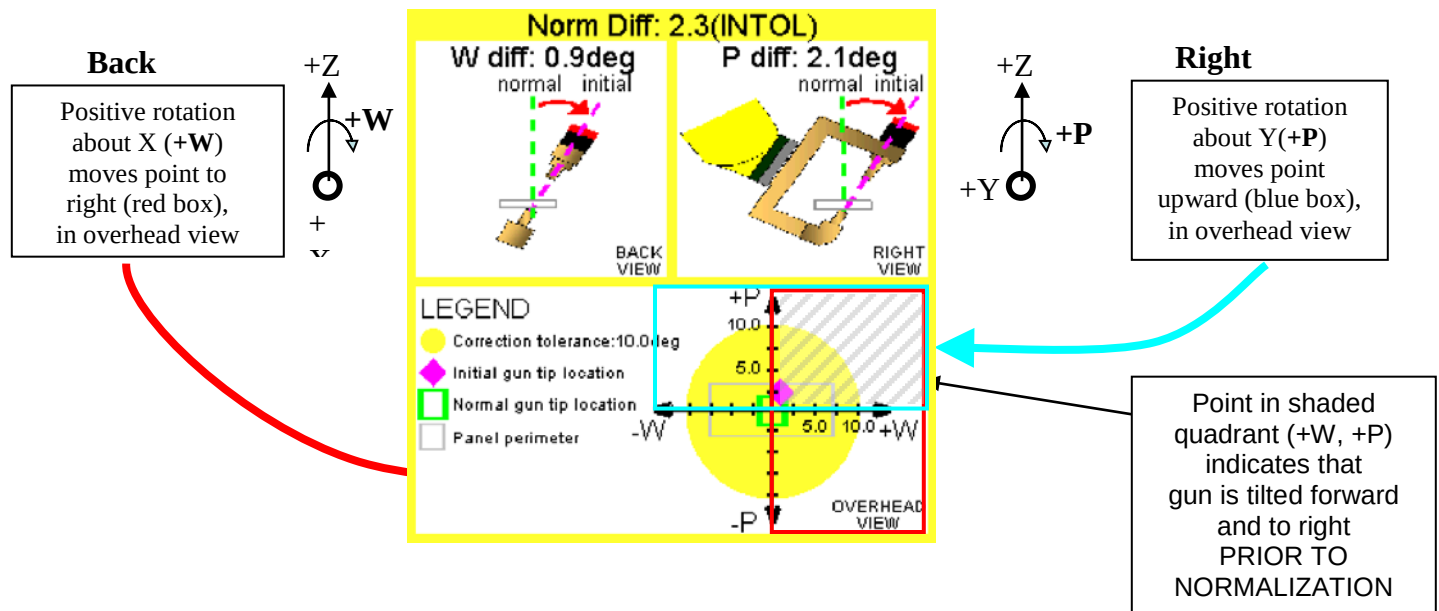
Table 4-1 Popup window persistence

Norm Diff	\$intol_ctlopt & 0x2 (0x2= Prompt for move to norm)	\$popupdwltm	Popup persistence
Intol	0	x(ms)	Popup dismissed after XX ms
Intol	1	*	Popup dismissed after user keypress
Overtol	*	*	Popup dismissed after user keypress

4.5. Explanation of Overhead view (Assuming Outdir = +x, Leftdir = +y)

Basically, overhead view provides direction perspective and correction tolerance.

From a physical perspective, overhead view is gun tip point viewed from a camera that is positioned above normal vector to taught point. In this perspective “normal vector is at center of view (origin), and correction tolerance is a circle with radius theta.



5

ADDITIONAL INFORMATION

5. ADDITIONAL INFORMATION

5.1. Complete Parameter List

Some normalize control parameters are displayed on normalize config screen, but most are not. Description and purpose of all parameters is provided here. Additional info for some parameters is at end of this list.

Some normalize control parameters are displayed on normalize config screen, but most are not. Description and purpose of all parameters is provided here. Additional info for some parameters is at end of this list.

Table 5-1

SG_NRMCFG_T		Default	Description
rot_err_max	float	10.0	Rotation detection tolerance (max error expected in path). Note this value determines search clearance. I.E. gun and robot tip must move further away from plane of expected surface, when rot_err_max increases (expectation of larger error).
rot_cor_tol	float	5.0	max correction applied without prompt
xy_loc_typ	SHORT	1	xy location of normal point (location at normalization completion). 1: where gun tip touches upper panel 2: where rbt tip touches upper panel
shape_typ	LONG	1	1: diamond, 2: triangle pointing out 3: triangle pointing in 4: line pointing leftward 5: line pointing outward
shape_rad	SHORT	4	distance between original pt and test pt
shape_radx	SHORT	5	Not used.
shape_rady	SHORT	5	Not used.
shape_opt	LONG	0	option map for shape control (0x1: independent radx, rady)
shape_spd	SHORT	200	Speed of motion TO test point (to reposition robot)
srch_spd	SHORT	4	Speed of motion AT test point (to detect panel)
srch_mod	SHORT	1	1 = DT (servo-based panel detection) 2 = DIN (external / sensor-based panel detection) 3 = hdi
pnldetdityp	SHORT	1	Panel detect input type for din search mode 1=din, 8=rdi, 14=tpin
pnldetdiidx	SHORT	0	Panel detect input index for din search mode
locn_ctlopt			
intolctlopt			Control options for point in tolerance: 0x1: mov home 1st, 0x2: prmpt 4 mov2nrm
ovrtolctlopt			Control options for point NOT in tolerance: 0x1: mov home 1st, 0x2: prmpt 4 mov2nrm
PS_tch_dst_mgn	SHORT	0	Not used.
tch_dst_mgn	float	1.0	margin for panel push at previous detection
z_mgn_rbt		5.0	
srch_rdy_mgn	float	2.0	search start distance margin, for premature detect check
srch_obsdst	float	4.0	distance for dt observation
nrm_dst_mgn	float	5.0	distance margin for guntip at normal pos

Table 5-1

varyrad4tst	LBOOLEAN	FALSE	test mode: increment search radius by rad_inc_val
rad_inc_val	SHORT	2	test mode: value of increment added to search radius
rad_strtval	SHORT	5	test mode: starting search radius
rad_endval	SHORT	5	test mode: ending search radius
PS_varyspd4tst	SHORT	0	not used
varyspd4tst	LBOOLEAN	FALSE	test mode: increment search radius by spd_inc_val
spd_inc_val	SHORT	1	test mode: value of increment added to search speed
spd_strtval	SHORT	4	test mode: starting search speed
spd_endval	SHORT	4	test mode: ending search speed
num_iters	SHORT	1	test mode: number of times to execute normalize sequence
cfg_opt_msk	LONG	0	not used
ztol14midpt	float	2.5	Tolerance for z difference (check) at segment1 midpt
ztol24midpt	float	2.5	Tolerance for z difference (check) at segment2 midpt
extra_long1	LONG	0	future use
extra_long2	LONG	0	future use
extra_real1	float	0.0	future use
extra_real2	float	0.0	future use
popupwin_enb	SHORT	TRUE	Enable popup window feature for result display.
popupwin_dwl	SHORT	2500	Duration (persistence) of popup window.
popupblnktm	SHORT	175	Duration (persistence) of flashing number during search.
lnlm_nrmdbg	SHORT	0	Maximum number of lines in debug log. File is not created, if sz <= 0
lnlmnrhmhist	SHORT	2	Maximum number of lines in history log. File is not created, if sz <= 0
lnlmprmeval	SHORT	2	Maximum number of lines in summary log. File is not created, if sz <= 0
tol4rot0img	float	0.25	Tolerance to display normal image in back view, right view. If Norm Diff <= Tol4rot0img, then normal image displays., otherwise graphic is shown with gun tilted to right or left.
sho_ovrhdvw	Boolean	TRUE	Display Overhead view in popup panel. If sho_ovrhdvw == FALSE, then only back and right view panels display.

6

ALARMS

6. ALARMS

SVGN	Message (text)	Sev	Cause	Remedy
277	OpenLim too small:%.1fmm required.	warn	Gun open distance required for normalization search is larger than open limit. Normalization process aborted because of this condition.	Initiate normalization process with smaller gun open distance or increase gun open limit (if possible).
351	Nrm srch input error	warn	Normalize search input DIN is invalid.	Map normalize search input to a valid input index via \$SGNRMCFG[].\$PNLDETDITYP and \$SGNRMCFG[].\$PNLDETDIIDX.
352	Static tqlim ctrl required	warn	Torque limit control for upper panel detection is not equal to static type2. Torque limit Type 2 usually provides best performance for normalization function. This message is only a recommendation – it is not a fault problem.	Go to torque limit control in teach setup screen. Select torque control type 2. Perform calibration.
353	Vx.midpt-Vy.midpt overtol(%2f)	warn	gun open distance should be identical for vx.midpt and vy.midpt. Present difference exceeds tolerance, so one or more of the test points may be incorrect. See SVGN-353,354,355 detail on next page.	Confirm that detection result is correct. Redo tqlimit calibration if misdetection occurred. Increase ztol14midpt if necessary.
354	Vx.midpt-Pt0 overtol(%2f)	warn	gun open distance @vx.midpt should be same as pt0. Present difference exceeds tolerance, so one or more of the test points may be incorrect. See SVGN-353,354,355 detail on next page.	Confirm that detection result is correct. Redo tqlimit calibration if misdetection occurred. Increase ztol14midpt if necessary.
355	Vy.midt-Pt0 overtol(%2f)	warn	gun open distance @vy.midpt should be same as pt0. Present difference exceeds tolerance, so one or more of the test points may be incorrect. See SVGN-353,354,355 detail on next page.	Confirm that detection result is correct. Redo tqlimit calibration if misdetection occurred. Increase ztol24midpt if necessary
356	Rot detection tol is too small	warn	Panel detection occurred sooner than expected (orange block / search ready margin in search segment detail drawing). This means that max rotation error is too small for the rotation error that actually exists in the path. Panel misdetection can also cause this alarm.	Confirm that detection result is correct. Redo tqlimit calibration if misdetection occurred.
357	W+=%03.1f->%03.1f, P+=%03.1f->%03.1f, R+=%03.1f->%03.1f	warn	Rotation change: org angle to normalized angle.	Status message – no corrective action required.
358	x+=%03.1f, y+=%03.1f, z+=%03.1f	warn	X,Y coordinate change: org pos to normalized pos.	Status message – no corrective action required.
359	Invalid before nrm user prog.	warn	B4nrmprgenb = TRUE and B4NRMPRGNM does not exist or is invalid.	Set B4nrmprgenb = FALSE or create valid program with specified name.
360	Invalid after nrm user prog.	warn	AFnrmprgenb = TRUE and AFNRMPRGNM does not exist or is invalid.	Set AFnrmprgenb = FALSE or create valid program with specified name.
361	Popup windows disabled.	warn	Popup window files (xml) could not be found, so normalize popup feature was disabled. Note this message is posted only at power-up (one time).	Refer to cause code
362	Normalize process failed.	warn	Normalization process did not complete. Refer to cause code for reason.	Refer to cause code
363	Invalid rbt close direction.	warn	Robot close direction is not set to x,y or z. I.E. it is set to fix or an invalid direction.	set robot clos direction to appropriate direction: +/-x, +/-y or +/-Z
364	Invalid gun out direction.	warn	Gun is not set to x,y or z OR it is set to the same direction as robot close direction.	set gun out direction to appropriate direction: +/-x, +/-y or +/-Z
365	Number of iterations is invalid	warn	Number of iterations < 1	Set \$SGNRMCFG[]\$.NUM_ITER > 0.
366	Speed range is invalid.	warn	Specified speed range is invalid. \$SGNRMCFG.\$SPD_ENDVAL>=\$SGNRMCFG.\$SPD_STRTVAL.	Set \$SGNRMCFG.\$SPD_STRTVAL>0, and set \$SGNRMCFG.\$SPD_ENDVAL>=\$SGNRMCFG.\$SPD_STRTVAL."
367	Radius range is invalid.	warn	Specified radius range is invalid. \$SGNRMCFG.\$SPD_ENDVAL>=\$SGNRMCFG.\$SPD_STRTVAL."	Set \$SGNRMCFG.\$RAD_STRTVAL>0, and set \$SGNRMCFG.\$RAD_ENDVAL>=\$SGNRMCFG.\$RAD_STRTVAL."
368	Speed increment is invalid.	warn	Specified speed increment is invalid.	Set \$SGNRMCFG.\$SPD_INCVAL>0,
369	Radius increment is invalid.	warn	Specified radius increment is invalid.	Set \$SGNRMCFG.\$RAD_INCVAL>0,
384	Invalid tool frame	stop	Normalization supports UT frame type only.	

7

OPERATION ON VIRTUAL ROBOT

7. OPERATION ON VIRTUAL ROBOT

- Set \$sggun#.\$steach.\$pnldflctdst > 0
- The popup win recall function is invoked by pressing SHIFT+Prev once on FRVRC, - screen displays for duration specified in persist time
- The toggle normalize function is not available because it is not possible to press prev + f3 simultaneously
- Set \$sgnrmcfg[].\$simang_out to set angle w, \$sgnrmcfg[].\$simang_lft to set angle p